SPARKY'S SCIENCE FACTS

What Is an Electric Circuit?

Electricity · Ages 7+

> **Short answer:** A circuit is a complete loop that electricity flows around, from a battery, through wires and a bulb, and back to the battery.

Sparky always says: Never play with plug sockets or wall electricity - it can kill. Only use small batteries for experiments.

How it works

- The battery pushes electricity around the loop. The bulb lights up when electricity flows through it.
- If there is a gap anywhere in the loop, the electricity stops and the bulb goes off.
- A switch is a gate: it closes the gap to turn the bulb on, and opens it to turn it off.
- Conductors let electricity flow: metals like copper and aluminium.
- Insulators stop electricity: plastic, rubber, wood and glass. Wires have a plastic coat to keep us safe.

Key words: Conductor: lets electricity through; Insulator: blocks electricity

Common mix-ups

- Electricity does not get used up in the wires: it flows around the loop.
- Water is not safe near electricity: tap water conducts, so never touch switches with wet hands.

Did you know? The plug sockets in South African homes give 230 volts. That is about 150 times more than one 1.5 volt battery.

Try it: With a grown-up, use a torch battery, two wires and a small bulb to test which things are conductors: a spoon, a pencil, a coin, an eraser.

Ask Sparky: Why does the bulb need two wires? Is the human body a conductor?